Database management systems and applications (ET -8201)

Department of Electronics and Telecommunication Engineering
Mbeya University of Science and Technology

Email : dsinkonde2013@gmail

Expected Outcomes

Students can expect to gain the following:

- ✓ Obtain basic concept in DBMS and the skills in designing simple databases
- ✓ Familiarize with different DBMS models
- ✓ Attain necessary skills to write queries in both SQL
- ✓ Analyze the contribution of database theory to practical implementations of database management systems.
- ✓ Interpret and discuss the impact of emerging database standards.

General Course Information

Faculty: **Daniel Sinkonde K,**

MUST Office: ETE No. 03

Phone: *0787/0754-952280*

E-mail: dsinkonde2013@gmail.com

Lectures: To be Communicated

Assignments and Learning Evaluation

- Mode of Assessment
- Continuous Assessment (CA) 40%

<u>Assignments</u>

- Total Score = 15% , End of submission on the 7th Week
  - 12th *December, 2023*
- **<u>Tests</u>**
- Test to be conducted Starting From Week 8 = 25%
  - *19th December, 2023*

- University Examination (UE) 60%

- Number of Credits : 9

Assignments

- Project
  a) Analysis of the literature
  b) Design of a solution
  c) Development of the designed solution
  d) Experimental Analysis
  e) Presentation Skills
- Different combinations for a, b, c, d, e but a+b+c+d+e= 15%
- Individual
- Knowledge of **Drawio** and **Microsoft Access**
- Topics available at the end of the slide (now :-)

General Course Information

Textbooks/References

1. Avi Silberschatz, Henry F. Korth , S. Sudarshan;(2011) Database System Concepts, 5th Edition.
2. Introduction to Oracle: SQL and PL/SQL version 1 & 2.
3. Ramez Elmasri, Shamkant B. Navathe; (2015), Fundamental of Database Systems, 6th Edition, Addison - Wesley, 20, ISBN: 978-013-608620-8.
4. Melton and Simon, (2016), Understanding the New SQL: A Complete Guide.

General Course Information

Textbooks/References

5. Tonny, R. (1995) GCSF Computer Studies, Letts Educational, London
6. Melton, Simon, and Gray, SQL: 1999 Understanding Relational Language Components. 2001.
7. Hoffer, J. A., Venkataraman, R., &Topi, H. (2010). Modern database management (10th ed.). Prentice Hall.
8. Pratt and Adamski, (2011), Database Systems Management and Design, 7th Edition, Course Technology.

Topic to be Covered

Introduction to computer:

Definition (database, DBMS), Example of different databases, Purpose of databases and their characteristics, History of database, Advantage of databases, File system in DBMS, file system versus database system, Database abstraction, database architecture, Database schema and instances, Database languages (DDL, SQL), Database users and administrators, Entity relationship Model (ER), Entity, attributes and attribute set, Entity types ,Entity set, Keys attributes, Relationship type and relationship set, Relationship cardinality, Naming, conversions and design issues, Notation Database schema.

Topic to be Covered

Database System Architectures:

Centralized and Client-Server Systems, Server System Architectures, Parallel Systems, Distributed Systems, Network Types.

Database Development Life Cycles/Processes communication: Planning (Feasibility study and Requirements gathering), Analysis (Identifying the entities, attributes, relationships, cardinalities and associations), Designing (Designing an E-R Diagram, Logical Schema, Functional Dependences and Normalization, Data dictionary and Physical Schema).

Topic to be Covered

Extended ER-design:

Subclass, super class, Inheritance, Specialization and generalization, Relation and relation constraints, Why relation, Mapping ER diagrams to Relations, entity set to relation, Relation constraints, Relation constraints examples.

Writing basic SQL Statements:

Capabilities of SQL SELECT Statements, Basic SELECT Statements, Writing SQL Statements, Selecting All columns, Selecting Specific Columns, Column Heading Defaults, Arithmetic Expressions, Using Arithmetic Operators, Operator Precedence, Using Parentheses, Defining a NULL Value, NULL VALUE in arithmetic Expressions, Defining a Column Alias, Using Column Alias, Concatenation Operation, Using the Concatenation Operator, Literal Character Strings, Using Literal Character Strings, Duplicate Rows, Eliminating Duplicate Rows.

Topic to be Covered

Restricting and Sorting Data:

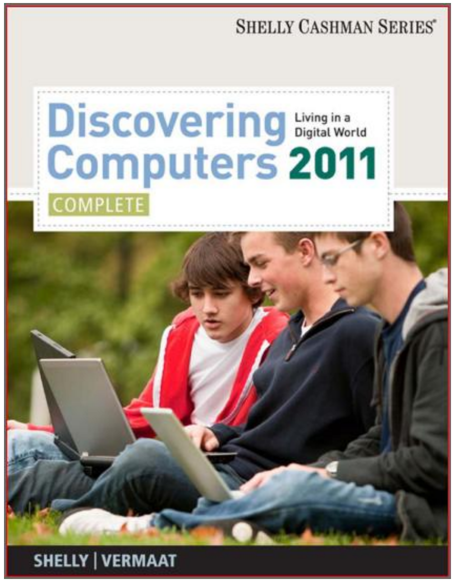
Limiting Rows Using a Selection, Limiting Rows Selected, Using the WHERE Clause, Character Strings and Dates, Comparison Operators, Using the Comparison Operations, Other Comparison Operators, Using the BWETWEEN Operator, Using the IN Operator, Using the LIKE Operator, Using the IS NULL Operator, Logical Operators, Using the AND Operator, Using the OR Operator, Using the NOT operator, Rules of Precedence, ORDER BY clause, Sorting in Descending Order, Sorting by Column Alias, Sorting by Multiple Columns.

Topic to be Covered

Single-Row Functions:

SQL Functions, Two Types of SQK Functions, Single-Row Functions, Character Functions, Case Conversion Functions, Using Case Conversion Functions, Character manipulation Functions, Using the Character Manipulation Functions, Number Functions, Using the ROUND Function, Using the TRUNC Function, Using the MOD Function, Working with Dates, Arithmetic with Dates, Using Arithmetic Operators with Date, Date Functions, Using Date Functions, Conversion Functions, Implicit Data type Conversion, Explicit Data type Conversion, TO_CHAR Function with Dates, Using TO_CHAR Function with Dates, TO_CHAR Function with Numbers, using TO_CHAR Function with numbers.

Lecture 1:



Email : dsinkonde2013@gmail

Common used in SQL

ORACLE DATABASE

MySQL

mongoDB

Microsoft Access

Microsoft SQL Server

MariaDB

PostgreSQL

IBM DB2

Microsoft Access 2007

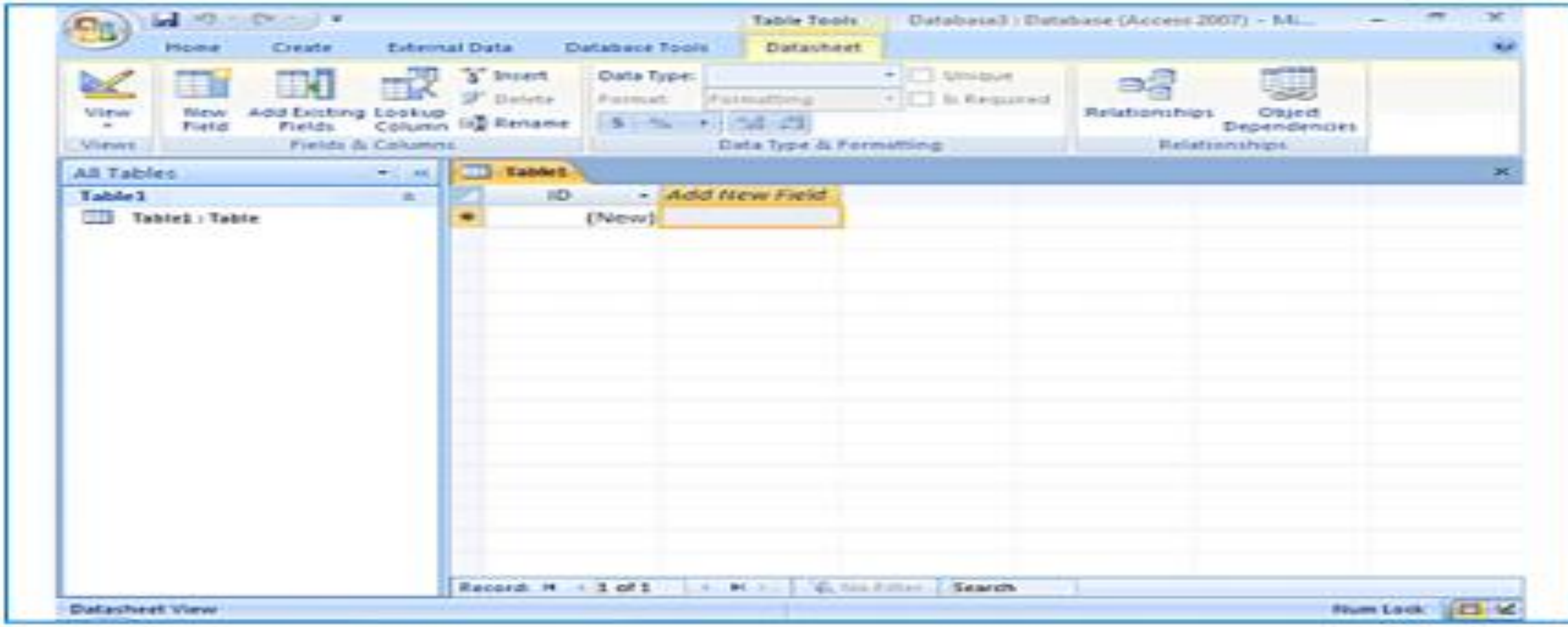
- **Introduction to Database Programs**

Lesson Objectives

After completing this lesson, you will be able to:

- Explain basic concepts of a database.
- Two different ways to create a database.
- Work with records in a database.
- Explain what database queries are and how they work.
- Work with reports

Overview of Access



A database contains objects that store, edit, and format information. Data is organized in a database in the form of tables.

Two common types of databases *Flat file* and *Relational.*

•**Flat file:** database contains all the data in a single table,

•**Relational:** database stores data in multiple tables.

Access Objects

Tables: - a collection of data about a topic arranged in rows and columns.

Forms: - a place to enter data easily

Queries: - a tool that lets you view, change, and analyze data in different ways

Reports: - a method to present your data in a printed format, such as charts & invoices

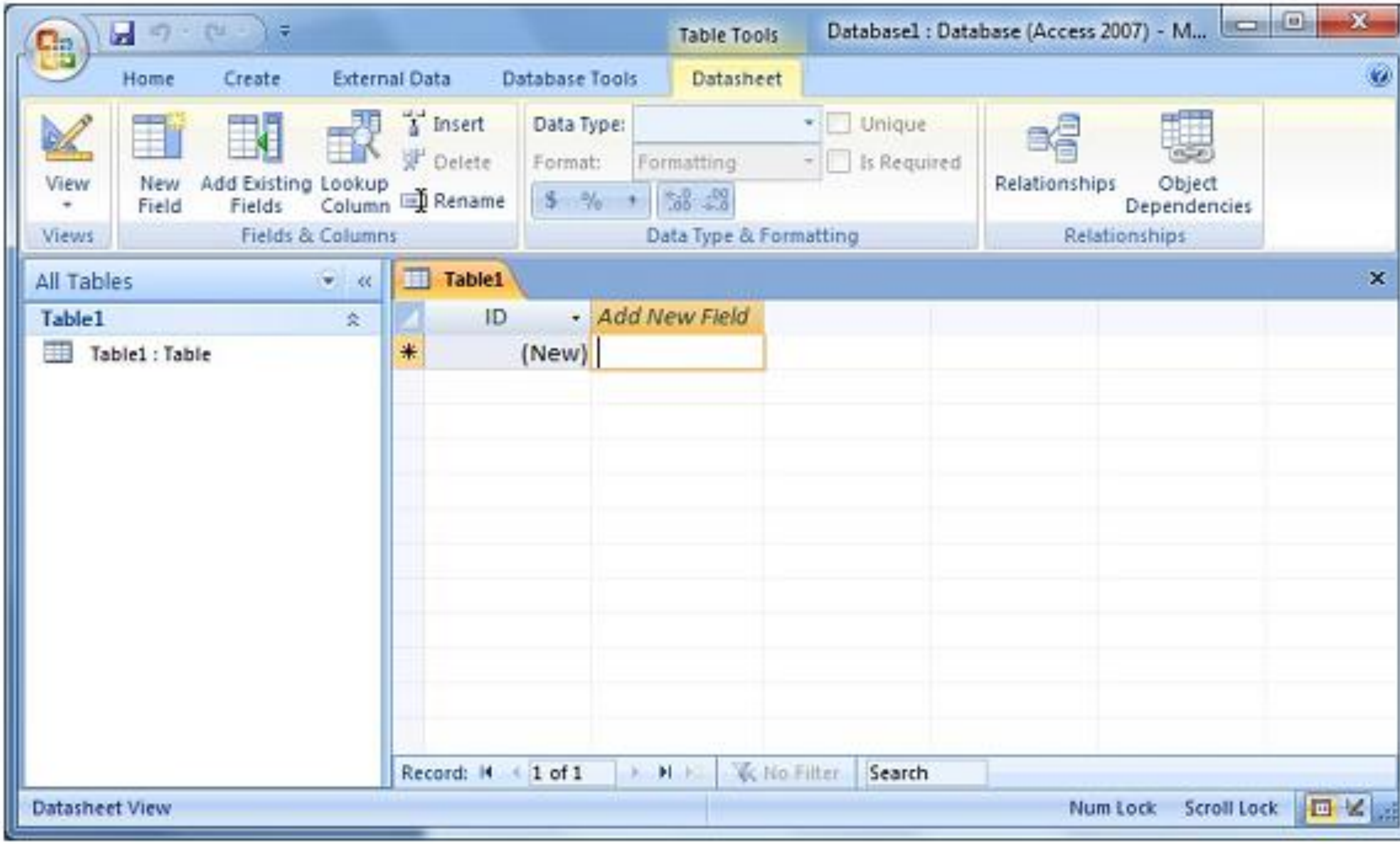
Create a Database

- click the **Start** button, and then click **All Programs**
- click **Microsoft Office**, and then click **Microsoft Office Access 2007**

Create a Database (cont.)

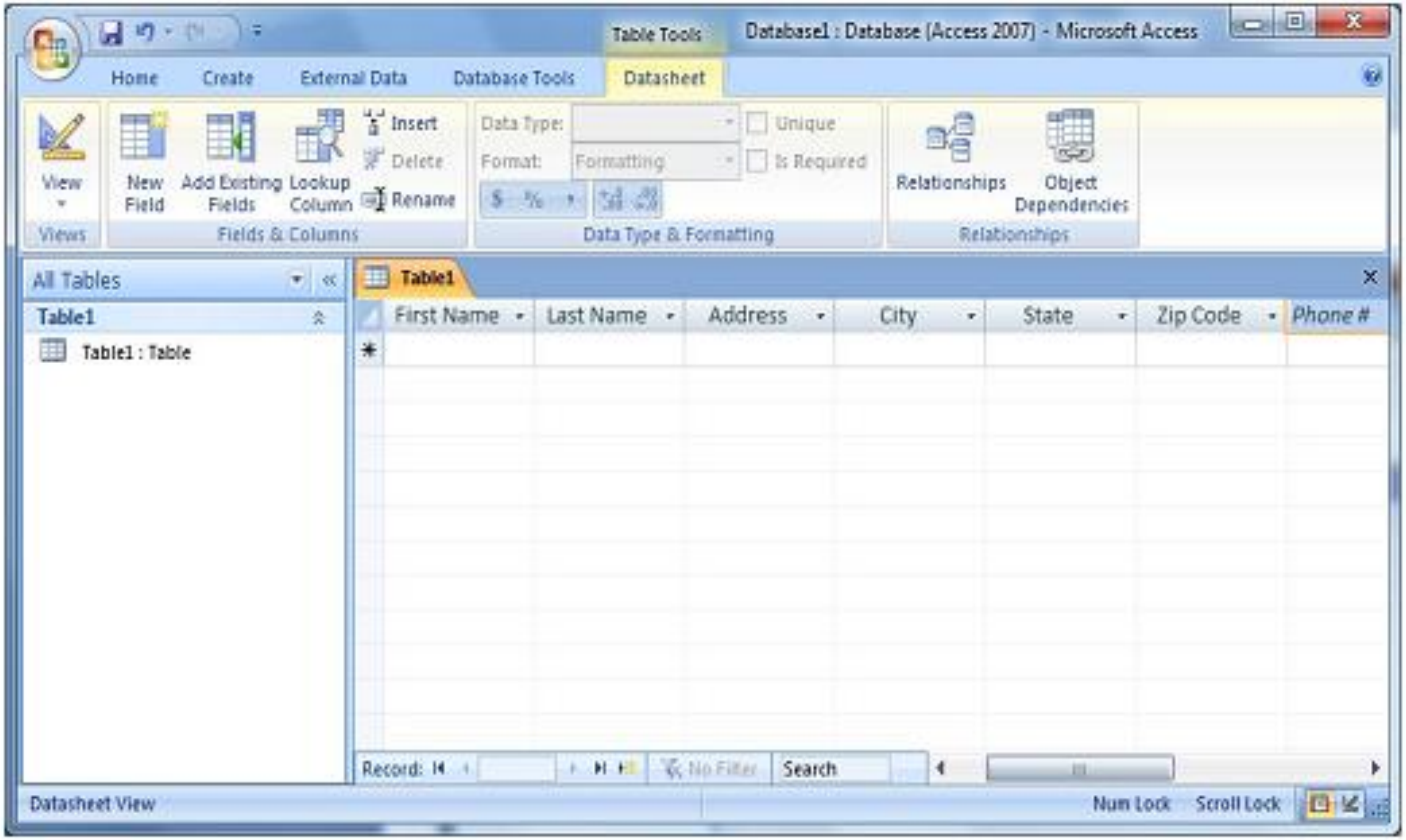
Click **Blank Database**. Type a name for your database in the File Name Box.

Create a Database (cont.)



To create fields for the table, double click the **Add New Field** cell, type First Name and press the ENTER key.

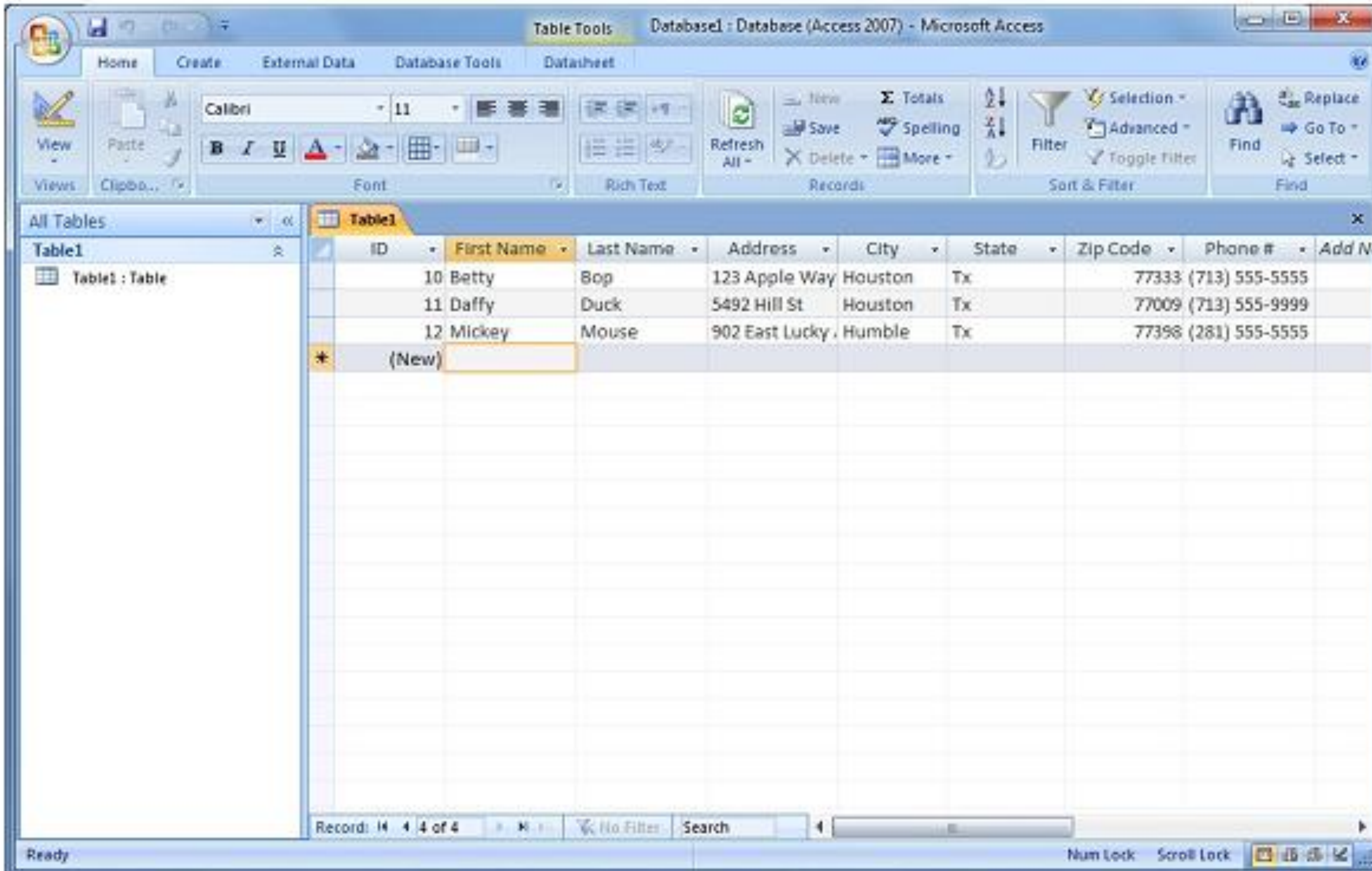
Create a Database (cont.)



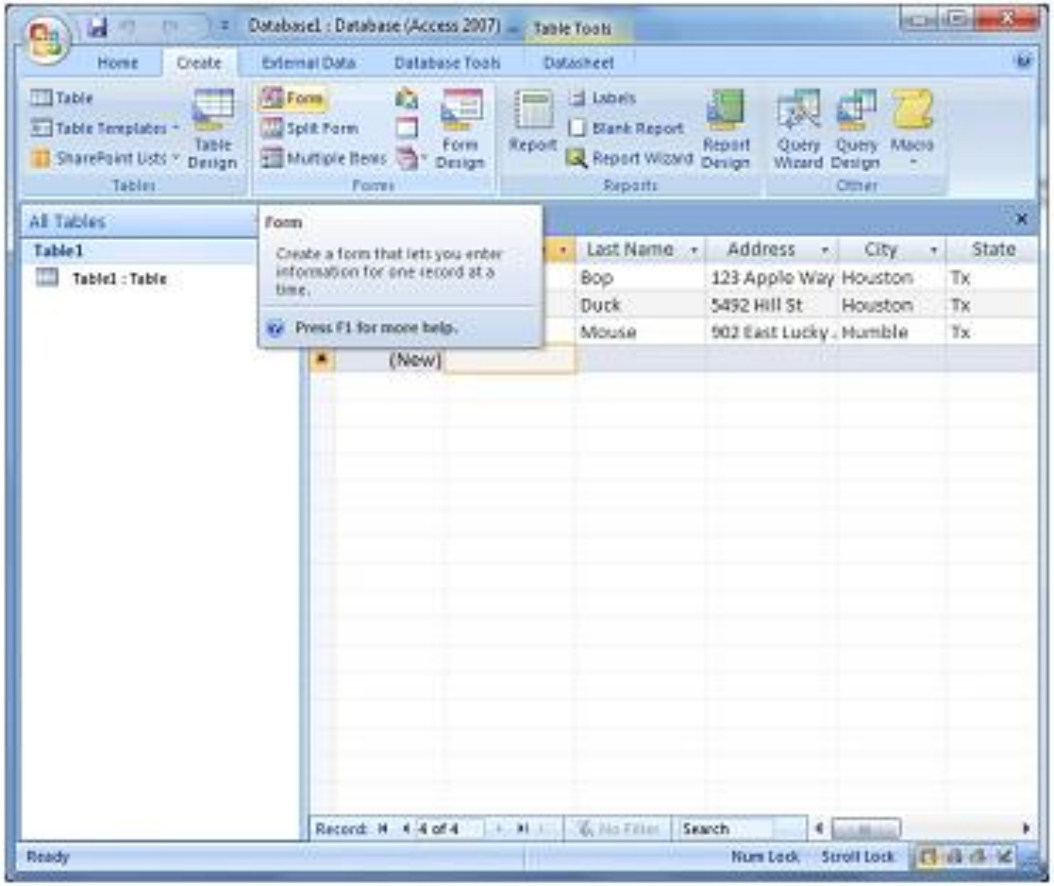
To add data to the table, type in the data in each field cell.

Create a Database (cont.)

Entering Data

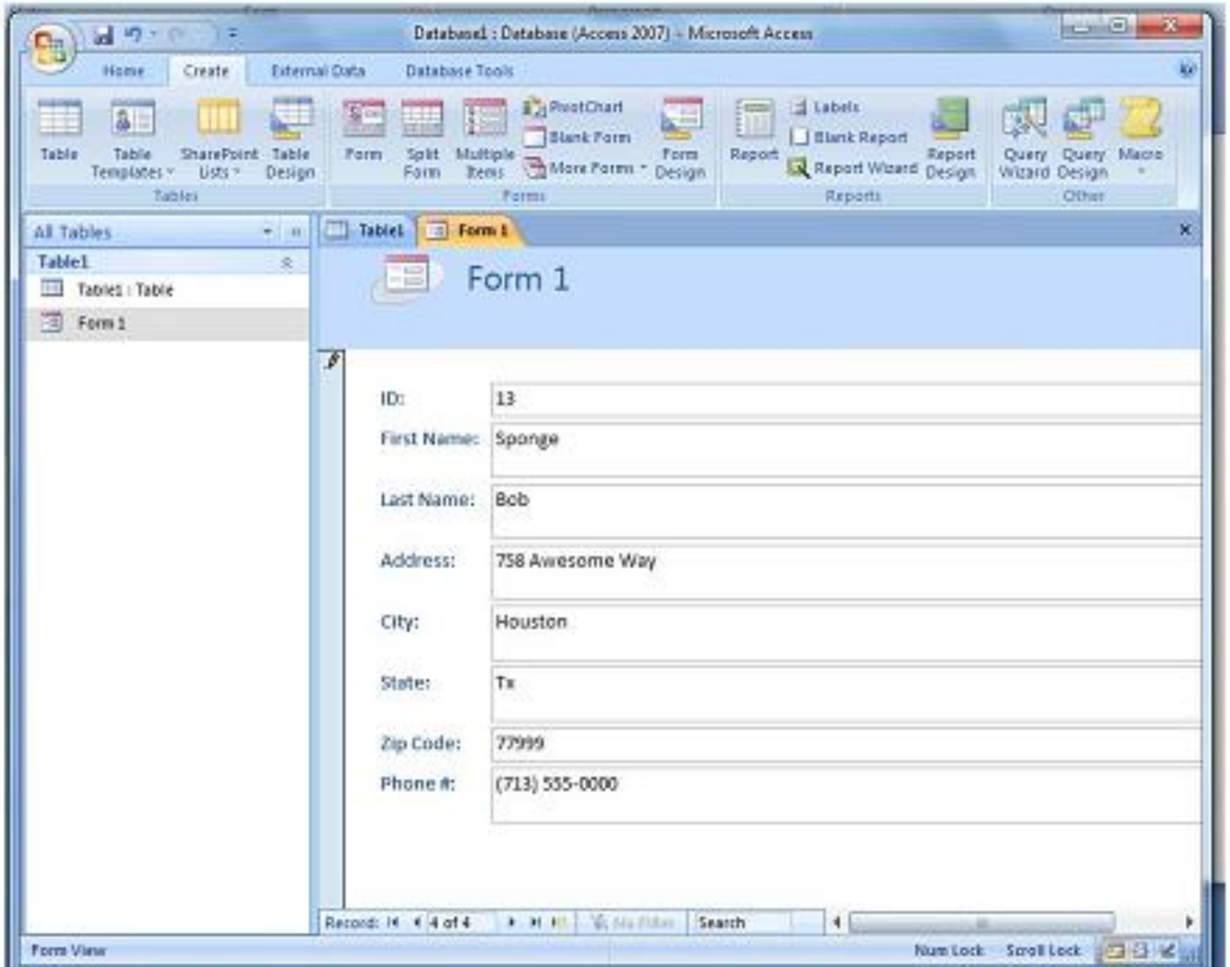


Creating Forms



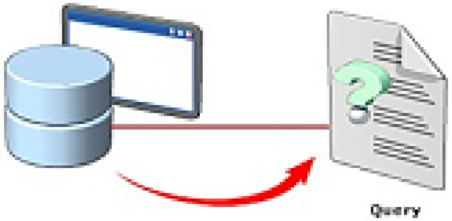
To create a form from a table, click on the Form icon from the Create tab.

Creating Forms (cont.)
Entering Data

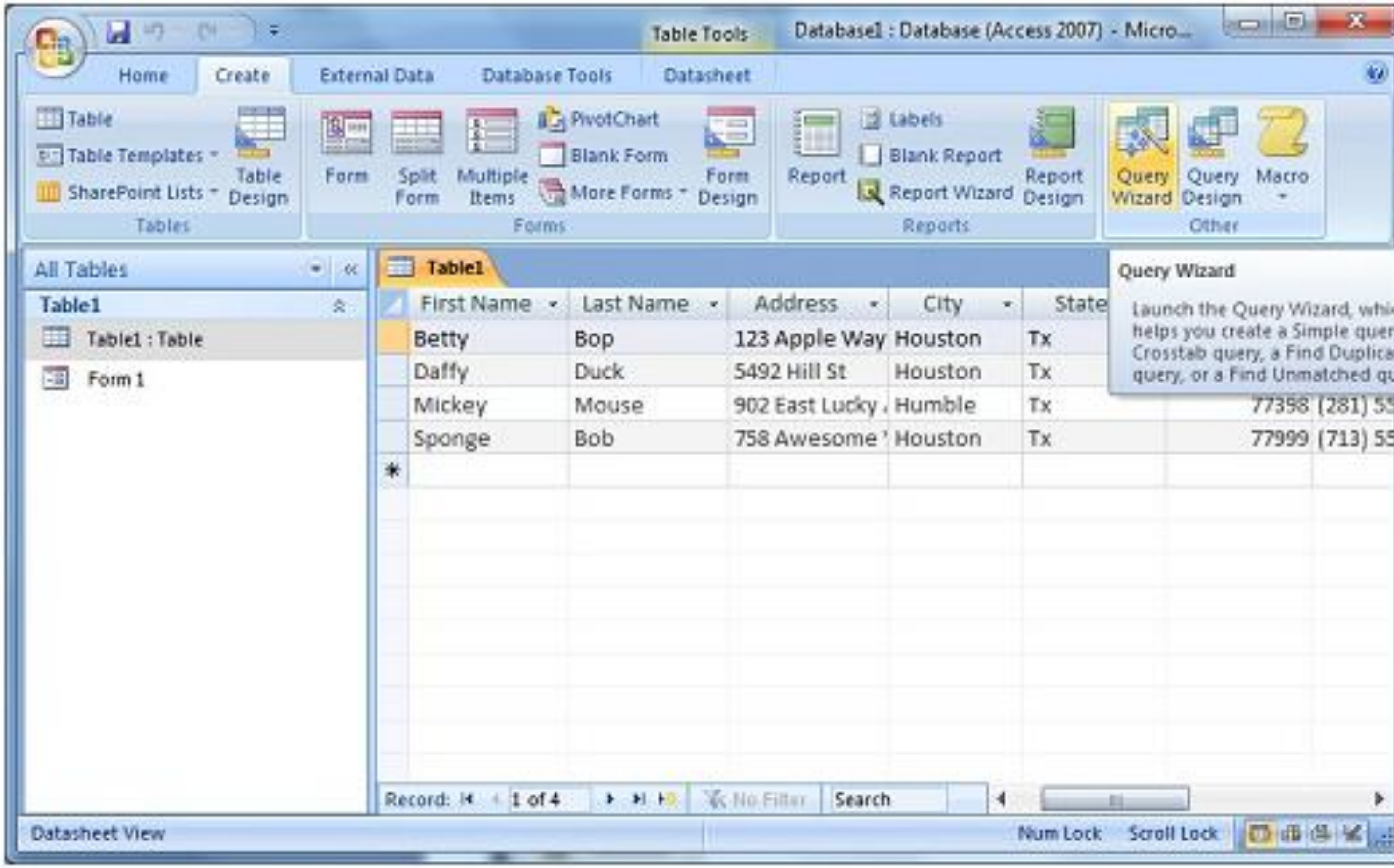


Database Queries

- You use a query to retrieve specific information from a database.
- A query is a question that you enter in a database program.
- The database program then performs the required operations to present the answer in the form of a report.
- A query helps you view specific data to modify or analyze it.

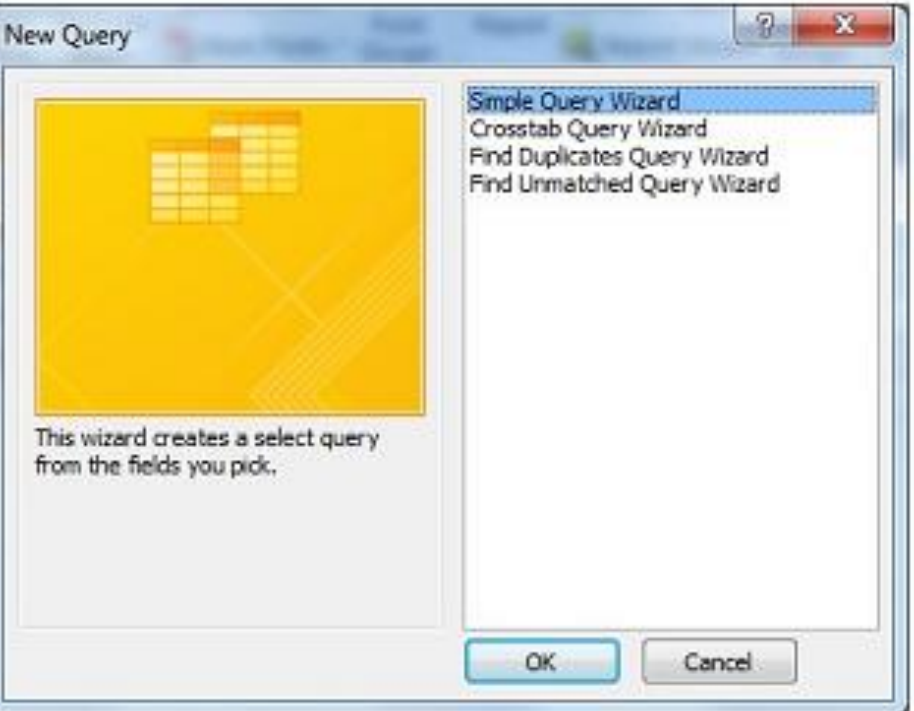


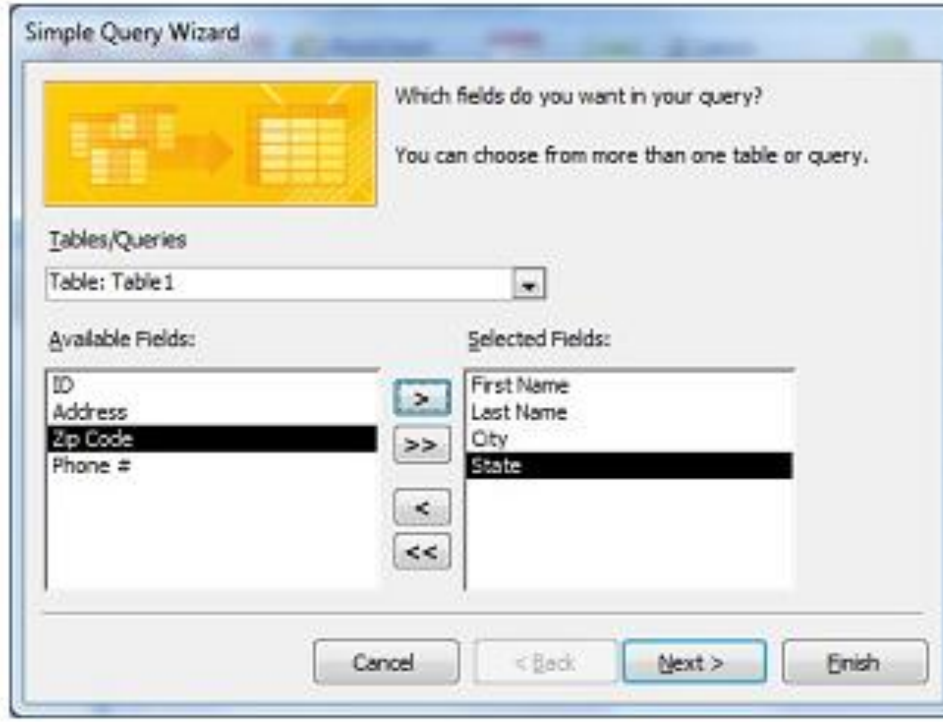
Creating Queries



To create a query, click the **Create** tab. In the Ribbon, click **Query Wizard**.

Creating Queries (cont.)

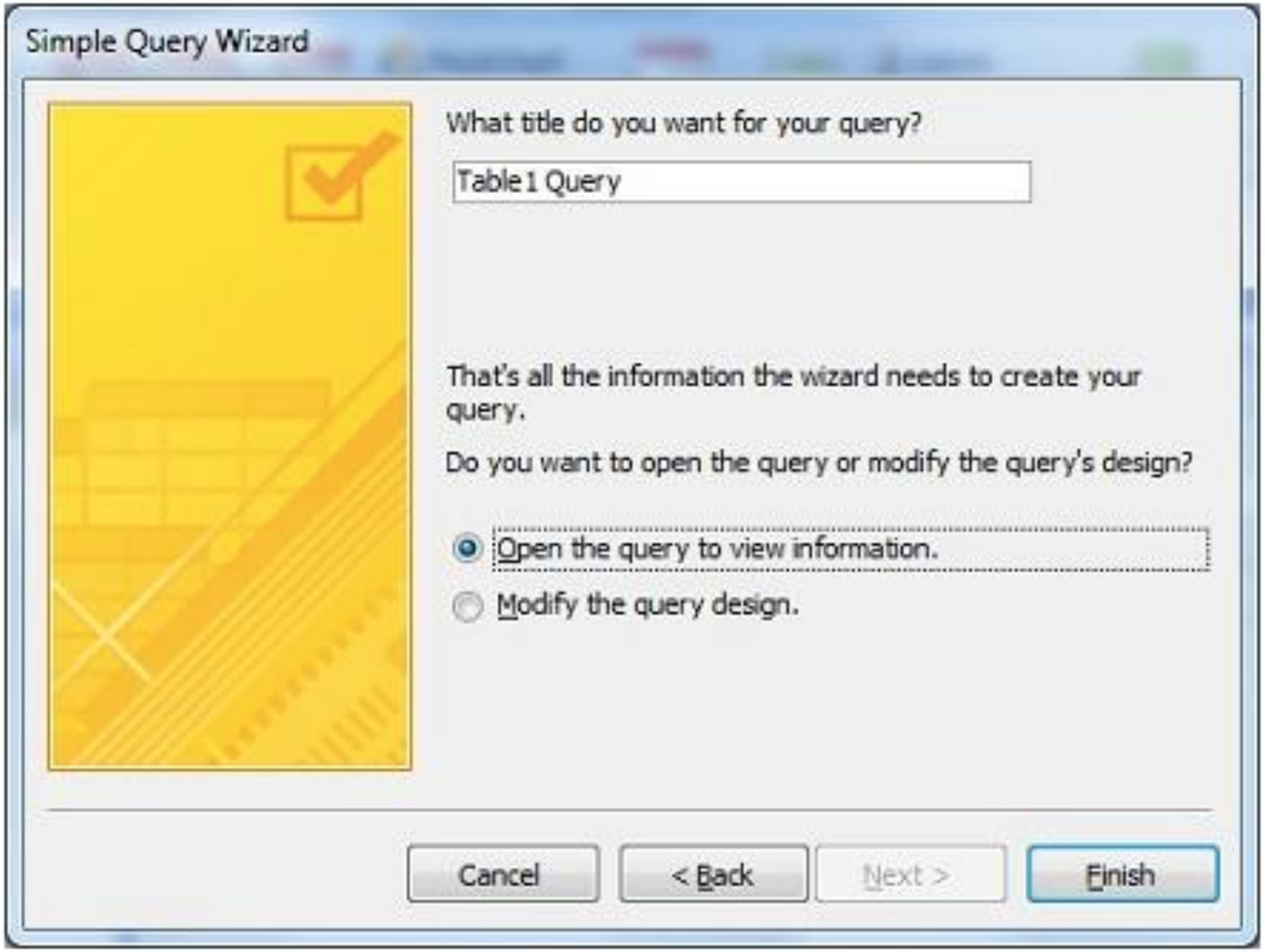




When the **New Query** window appears, make sure **Simple Query Wizard** is selected.

Transfer data from the Available Field to the Selected Field one at a time using >

Creating Queries (cont.)

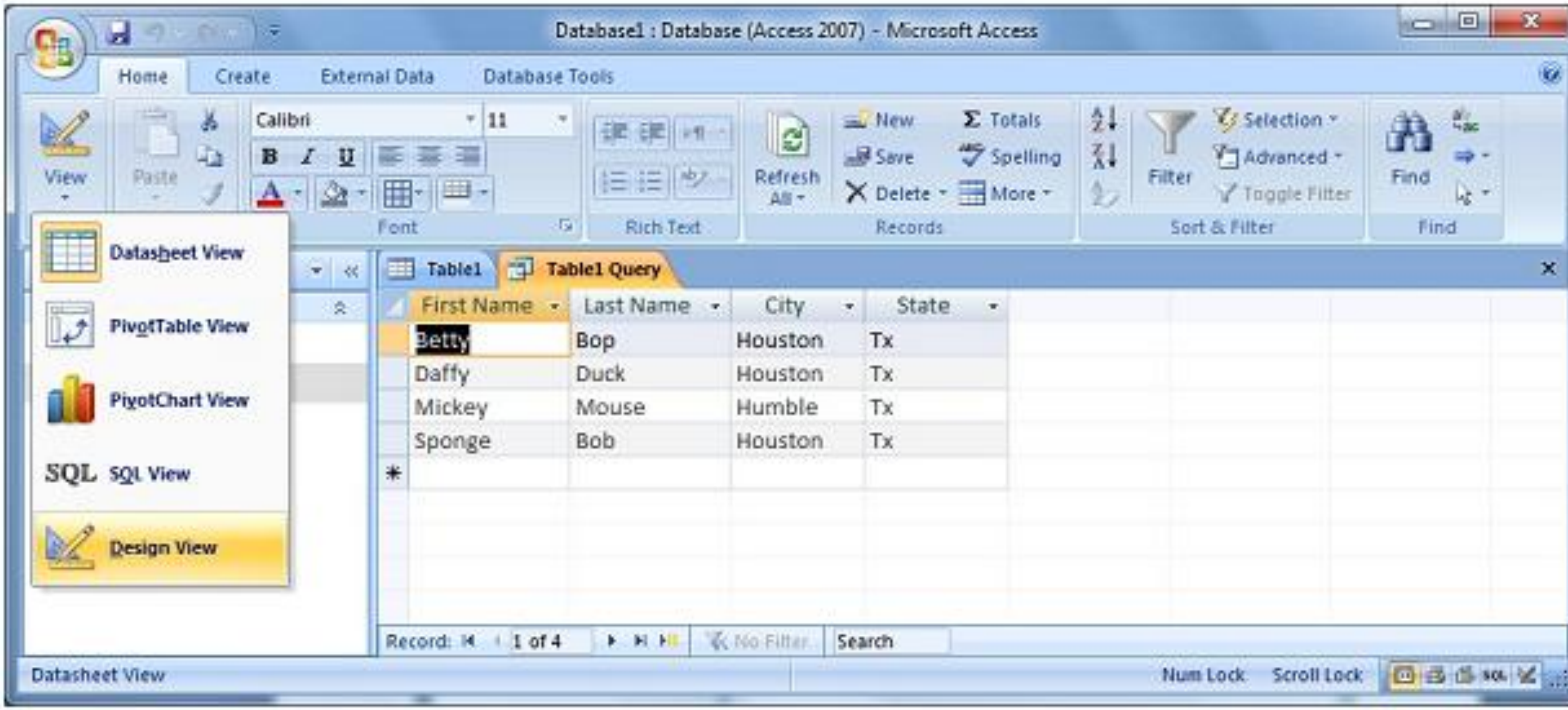


When the final screen appears, type the **Names of the query**

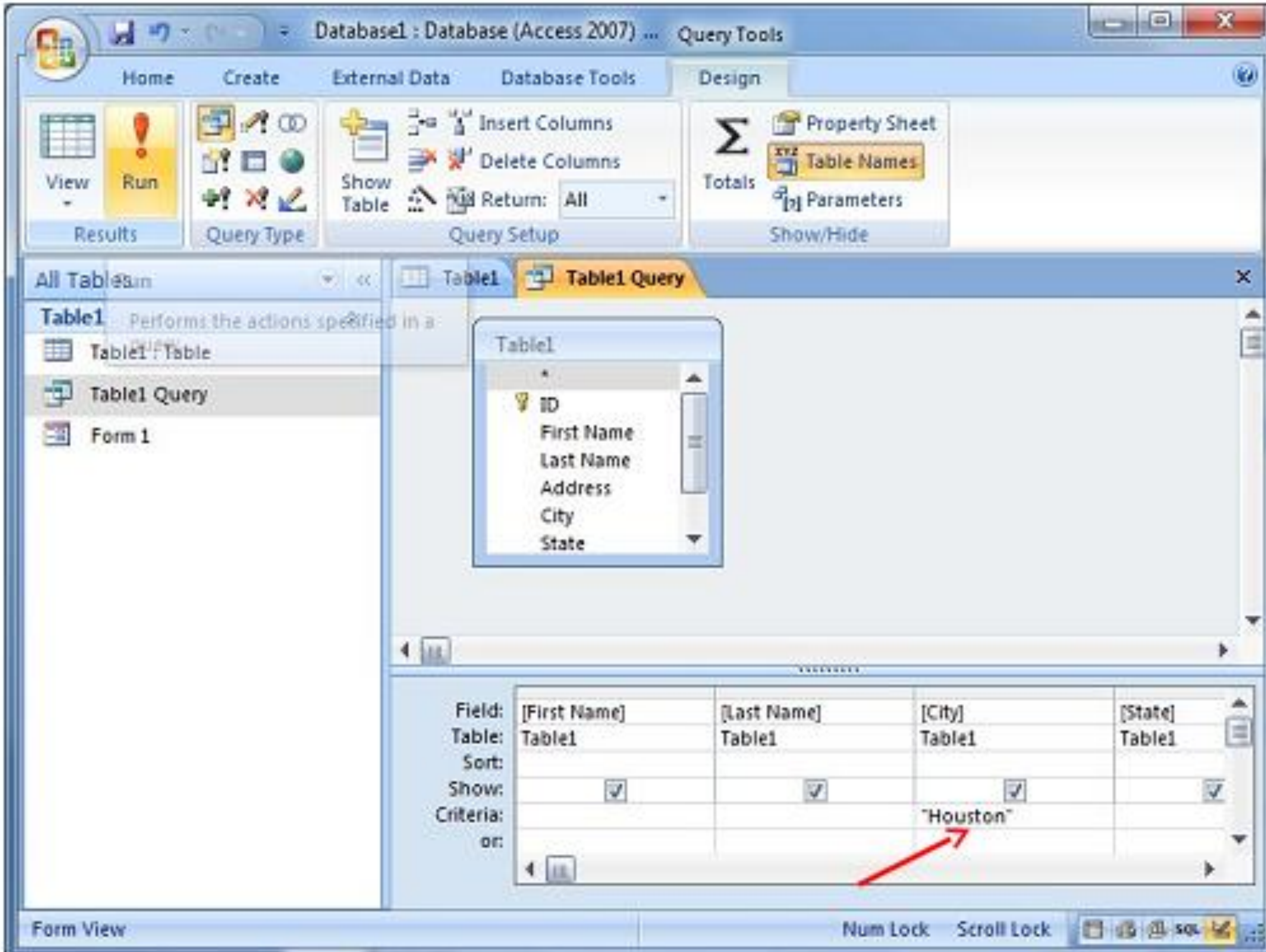
Creating Queries (cont.)

The query is automatically saved and executed.

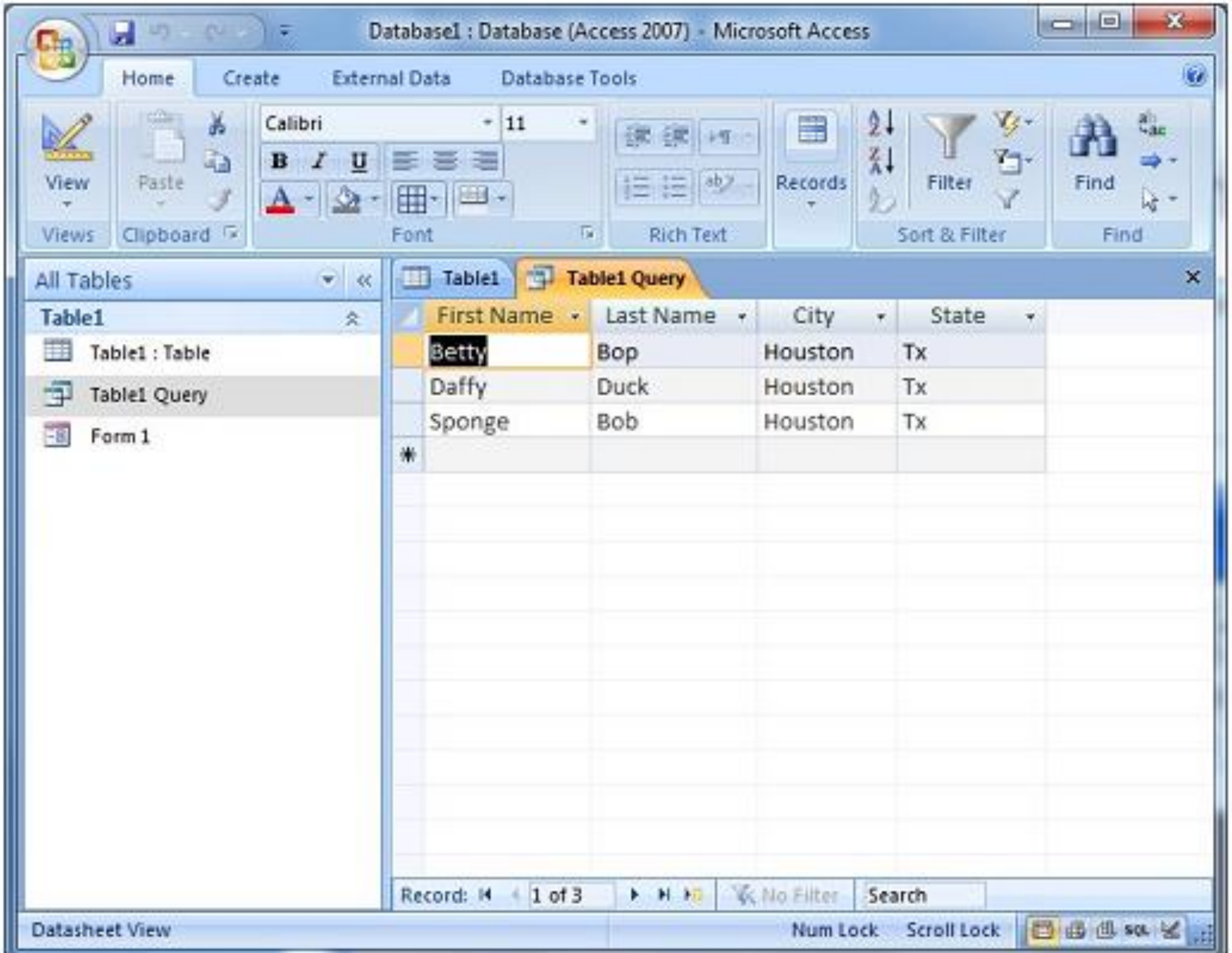
Creating Queries (cont.)



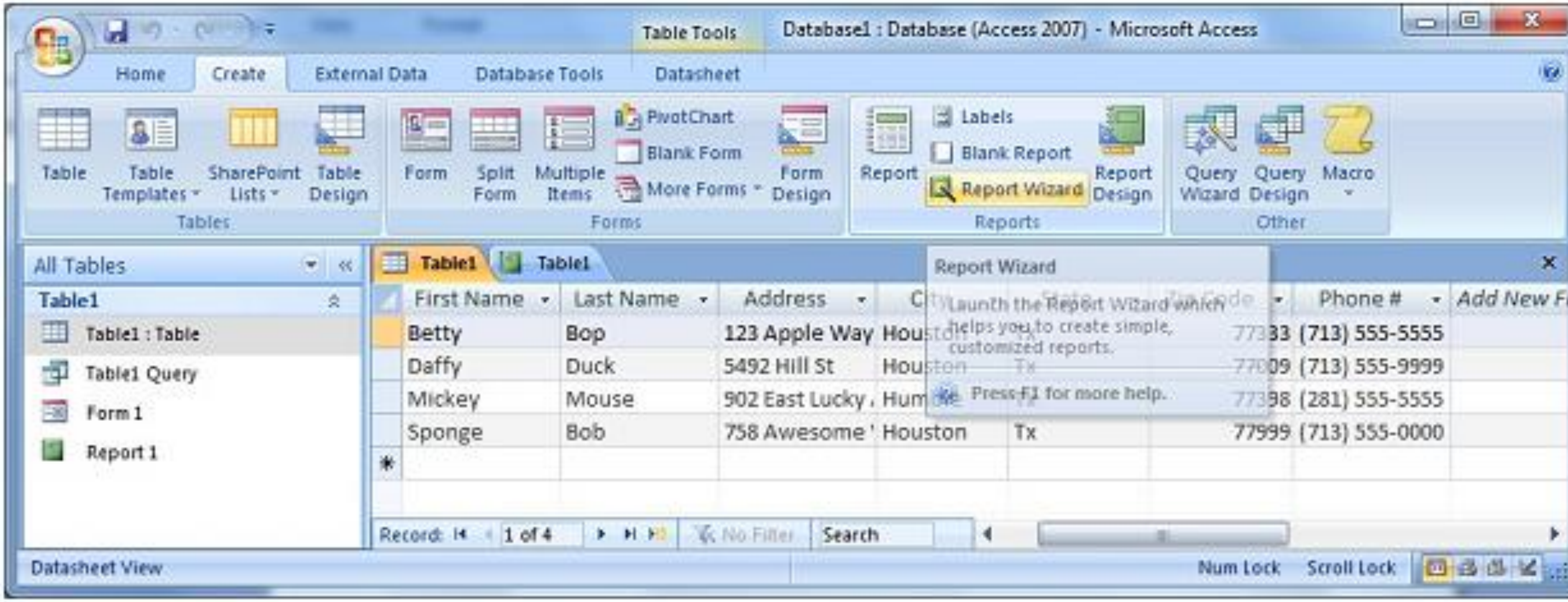
Creating Queries (cont.)



Creating Queries (cont.)

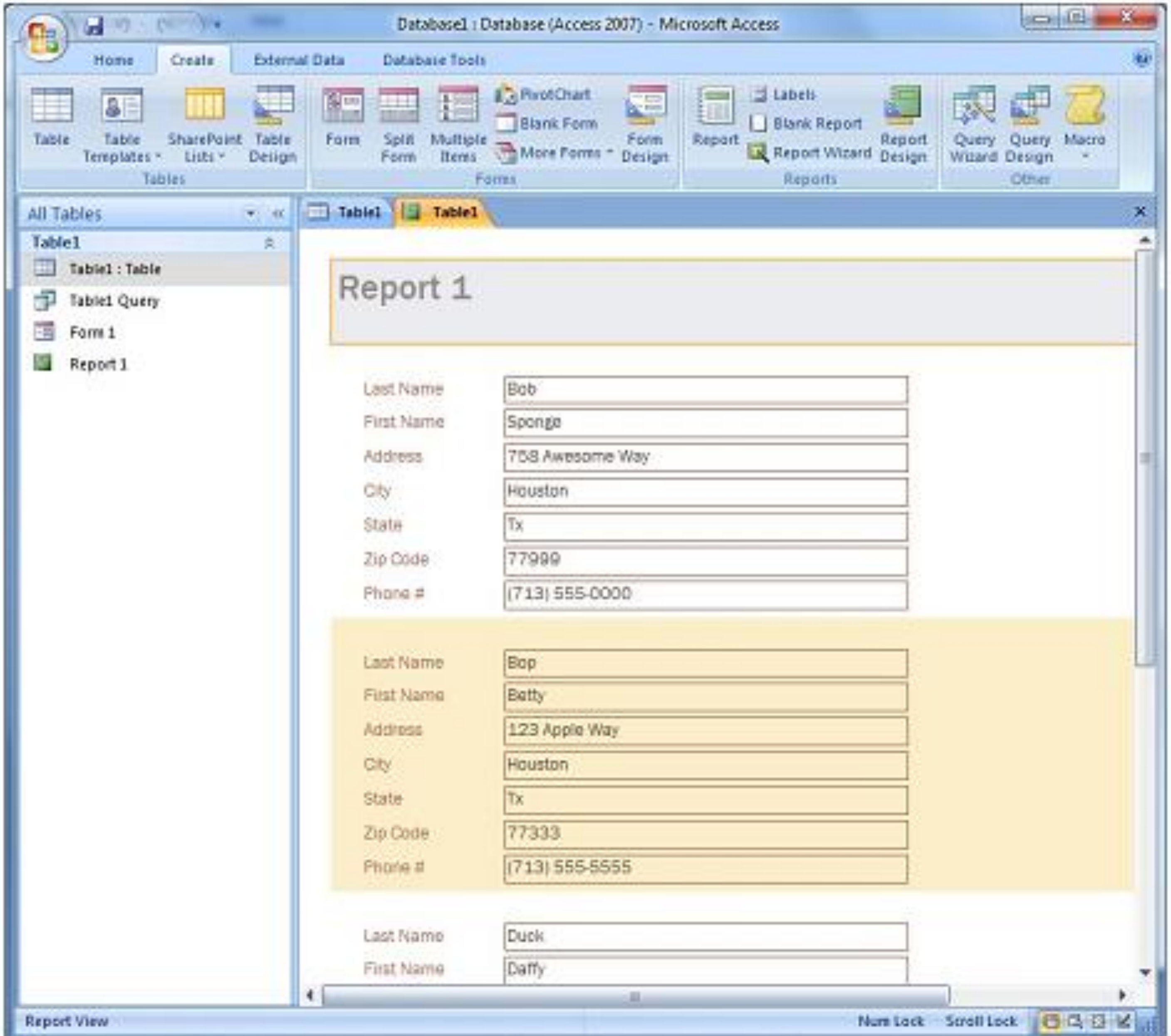


Working with Reports



To create a report by using the Report Wizard, click the **Create** tab, and in the **Reports** group, click **Report Wizard**.

Working with Reports



Questions

End of segment